

EECS 16A Foundations of Signals, Systems, & Information Processing

Fall 2025 Homework 5

This homework is due Friday, October 3, 2025, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. DTFS Expansion (Adapted from EE 120 Homework)

Determine the complex-exponential discrete-time Fourier series (DTFS) expansion for each signal $x: \mathbb{Z} \rightarrow \mathbb{R}$ described below, or explain why no such expansion exists. For each case where a DTFS expansion exists, be sure to identify the period p and the fundamental frequency ω_0 .

For part (d), note that we include the Kronecker-Delta function δ , which is defined as:

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$$

(a) $x[n] = \cos\left(\frac{3\pi}{5}n\right) + \frac{1}{3}\cos\left(\frac{4\pi}{5}n\right), \forall n \in \mathbb{Z}.$

Solution:

$$x[n] = \cos\left(\frac{3\pi}{5}n\right) + \frac{1}{3}\cos\left(\frac{4\pi}{5}n\right) = \frac{1}{2}(e^{i\frac{3\pi}{5}n} + e^{-i\frac{3\pi}{5}n}) + \frac{1}{6}(e^{i\frac{4\pi}{5}n} + e^{-i\frac{4\pi}{5}n}) = \frac{1}{6}e^{-i\frac{4\pi}{5}n} + \frac{1}{2}e^{-i\frac{3\pi}{5}n} + \frac{1}{2}e^{i\frac{3\pi}{5}n} + \frac{1}{6}e^{i\frac{4\pi}{5}n}$$

The fundamental period of a signal is the lowest common multiple of the fundamental periods of each of the exponential terms in the DTFS expansion. For our expansion in this part, the terms have the following fundamental periods:

- i. $\frac{1}{6}e^{-i\frac{4\pi}{5}n}$: Period $\frac{2\pi k}{\frac{4\pi}{5}} = \frac{10\pi k}{4\pi} = \frac{5k}{2} \rightarrow p = 5$
- ii. $\frac{1}{2}e^{-i\frac{3\pi}{5}n}$: Period $\frac{2\pi k}{\frac{3\pi}{5}} = \frac{10\pi k}{3\pi} = \frac{10k}{3} \rightarrow p = 10$
- iii. $\frac{1}{2}e^{i\frac{3\pi}{5}n}$: Period $\frac{2\pi k}{\frac{3\pi}{5}} = \frac{10\pi k}{3\pi} = \frac{10k}{3} \rightarrow p = 10$
- iv. $\frac{1}{6}e^{i\frac{4\pi}{5}n}$: Period $\frac{2\pi k}{\frac{4\pi}{5}} = \frac{10\pi k}{4\pi} = \frac{5k}{2} \rightarrow p = 5$

Therefore, as the least common multiple between 5 and 10, our fundamental period is $p = 10$. Correspondingly, our fundamental frequency is $\omega_0 = \frac{2\pi}{p} = \frac{\pi}{5}$. To find the Fourier coefficients X_k , we can apply the rule that X_k corresponds to frequency $k\omega_0$. Matching with our complex exponentials, we find that $X_{-4} = \frac{1}{6}, X_{-3} = \frac{1}{2}, X_3 = \frac{1}{2}, X_4 = \frac{1}{6}$, and all other coefficients are 0.

(b) $x[n] = \cos\left(\frac{\sqrt{2}\pi}{5}n\right), \forall n \in \mathbb{Z}.$

Solution:

$$x[n] = \cos\left(\frac{\sqrt{2}\pi}{5}n\right)$$

Therefore,

$$\frac{\sqrt{2}\pi}{5}p = 2k\pi \Leftrightarrow p = 5\sqrt{2}k$$

As a multiple of $\sqrt{2}$, which is not an integer, the period p cannot be an integer. This means that x is not periodic in discrete time, and so we cannot write a DTFS for it.

(c) $x[n] = \cos\left(\frac{2\pi}{3}n\right) + (-1)^n, \forall n \in \mathbb{Z}.$

Solution:

$$x[n] = \cos\left(\frac{2\pi}{3}n\right) + (-1)^n = \cos\left(\frac{2\pi}{3}n\right) + e^{i\pi n} = \frac{1}{2}(e^{i\frac{2\pi}{3}n} + e^{-i\frac{2\pi}{3}n}) + e^{i\pi n} = \frac{1}{2}e^{-i\frac{2\pi}{3}n} + \frac{1}{2}e^{i\frac{2\pi}{3}n} + e^{i\pi n}$$

The fundamental period of a signal is the lowest common multiple of the fundamental periods of each of the exponential terms in the DTFS expansion. For our expansion in this part, the terms have the following fundamental periods:

- i. $\frac{1}{2}e^{-i\frac{2\pi}{3}n}$: Period $\frac{2\pi k}{\frac{2\pi}{3}} = \frac{6\pi k}{2\pi} = 3k \rightarrow p = 3$
- ii. $\frac{1}{2}e^{i\frac{2\pi}{3}n}$: Period $\frac{2\pi ki}{\frac{2\pi}{3}} = \frac{6\pi k}{2\pi} = 3k \rightarrow p = 3$
- iii. $e^{i\pi n}$: Period $\frac{2\pi k}{\pi} = 2k \rightarrow p = 2$

Therefore, as the least common multiple between 2 and 3, our fundamental period is $p = 6$. Correspondingly, our fundamental frequency is $\omega_0 = \frac{2\pi}{p} = \frac{\pi}{3}$. Similar to part (a), we can find the fourier coefficients by matching with our complex exponentials: $X_{-2} = \frac{1}{2}, X_2 = \frac{1}{2}, X_3 = 1$, and all other coefficients are 0.

(d) $x[n] = \sum_{l=-\infty}^{\infty} \delta[n - lp],$ where p is a positive integer.

Solution:

We observe that our function $x[n]$ can alternatively be described as follows:

$$x[n] = \begin{cases} 1 & \text{if } n \text{ is a multiple of } p \\ 0 & \text{otherwise} \end{cases}$$

Observing this pattern, we can see that $x[n]$ is p -periodic, repeating a single value of 1 every p timesteps before continuing to output 0.

To find the Fourier series coefficients for this function, we can use the analysis equation across one period, described in the following general form:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}$$

Plugging in our definition for $x[n]$, we get:

$$\begin{aligned}
 X_k &= \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n} \\
 &= \frac{1}{p} \left(\sum_{n=0}^0 x[n] e^{-ik\omega_0 n} + \sum_{n=1}^{p-1} x[n] e^{-ik\omega_0 n} \right) \\
 &= \frac{1}{p} (x[0] e^{-ik\omega_0(0)} + \sum_{n=1}^{p-1} (0) e^{-ik\omega_0 n}) \\
 &= \frac{1}{p} ((1) e^0 + \sum_{n=1}^{p-1} (0)) \\
 &= \frac{1}{p} (1 + 0) = \frac{1}{p}
 \end{aligned}$$

Therefore, all of our Fourier series coefficients are $X_k = \frac{1}{p} \forall k$.

To write $x[n]$ as our DTFS expansion with these coefficients, we use our Synthesis equation:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} = \sum_{k=0}^{p-1} \frac{1}{p} e^{ik\omega_0 n} = \frac{1}{p} \sum_{k=0}^{p-1} e^{ik\omega_0 n}$$

As stated before, our fundamental period is p and our resulting fundamental frequency is $\omega_0 = \frac{2\pi}{p}$.

2. DTFS & DFT Properties

- (a) Let $\boldsymbol{\psi}_0, \dots, \boldsymbol{\psi}_{p-1}$ be the DTFS basis vectors. What is $\|\boldsymbol{\psi}_k\|, \forall k \in [0, p-1]$?

Hint: What is $\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_k \rangle$?

Solution: We know that:

$$\begin{aligned} \|\boldsymbol{\psi}_k\|^2 &= \langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_k \rangle = \boldsymbol{\psi}_k^T \boldsymbol{\psi}_k^* = \begin{bmatrix} e^{i\frac{2\pi}{p}(0)(k)} \\ e^{i\frac{2\pi}{p}(1)(k)} \\ \vdots \\ e^{i\frac{2\pi}{p}(p-1)(k)} \end{bmatrix}^T \begin{bmatrix} e^{-i\frac{2\pi}{p}(0)(k)} \\ e^{-i\frac{2\pi}{p}(1)(k)} \\ \vdots \\ e^{-i\frac{2\pi}{p}(p-1)(k)} \end{bmatrix} \\ &= \sum_{n=0}^{p-1} e^{i\frac{2\pi}{p}(n)(k)} e^{-i\frac{2\pi}{p}(n)(k)} = \sum_{n=0}^{p-1} e^0 = p \\ &\rightarrow \|\boldsymbol{\psi}_k\| = \sqrt{p} \end{aligned}$$

- (b) Show that DTFS is a *linear transformation*.

More formally defined, let a, b be scalars in $\mathbb{R}$, and x, y be discrete time periodic signals of length p (so, p -periodic). Let $f(s)$ define the DTFS representation of a given periodic signal s .

Show that $f(ax + by) = af(x) + bf(y)$.

Solution: Let $\mathbf{x}$ be the vector representation of x , and $\mathbf{y}$ be the vector representation of y .

$$\begin{aligned} f(ax + by) &= \frac{1}{p} \begin{bmatrix} \langle a\mathbf{x} + b\mathbf{y}, \boldsymbol{\psi}_0 \rangle \\ \vdots \\ \langle a\mathbf{x} + b\mathbf{y}, \boldsymbol{\psi}_{p-1} \rangle \end{bmatrix} = \frac{1}{p} \begin{bmatrix} \langle a\mathbf{x}, \boldsymbol{\psi}_0 \rangle \\ \vdots \\ \langle a\mathbf{x}, \boldsymbol{\psi}_{p-1} \rangle \end{bmatrix} + \frac{1}{p} \begin{bmatrix} \langle b\mathbf{y}, \boldsymbol{\psi}_0 \rangle \\ \vdots \\ \langle b\mathbf{y}, \boldsymbol{\psi}_{p-1} \rangle \end{bmatrix} = a \frac{1}{p} \begin{bmatrix} \langle \mathbf{x}, \boldsymbol{\psi}_0 \rangle \\ \vdots \\ \langle \mathbf{x}, \boldsymbol{\psi}_{p-1} \rangle \end{bmatrix} + b \frac{1}{p} \begin{bmatrix} \langle \mathbf{y}, \boldsymbol{\psi}_0 \rangle \\ \vdots \\ \langle \mathbf{y}, \boldsymbol{\psi}_{p-1} \rangle \end{bmatrix} \\ &= af(x) + bf(y) \end{aligned}$$

- (c) Recall that the DFT has the DTFS basis vectors scaled by $\frac{1}{p}$. Is it true that the DFT is also a linear transformation? Briefly justify your answer using your answer to part (b).

Solution: Yes! It is also a linear transformation since scaling every basis vector by a constant does not break linearity. Thus, scaling the DTFS basis vectors to get our DFT basis preserves its property of being a linear transformation.

Thus, DFT is a linear transformation.

You can formally prove this using the exact same procedure and work as part (b), just using the DFT basis vectors, rather than the DTFS basis.

3. Mystery DTFS (Adapted from EE 120 Discussion)

Consider a periodic real-valued discrete-time signal x having fundamental period p and an exponential discrete Fourier series expansion

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad (1)$$

where ω_0 is the fundamental frequency of the signal. The signal has the following properties:

- (i) $x[n+3] = x[n], \forall n \in \mathbb{Z}$
- (ii) $\sum_{n=0}^{p-1} x[n] = 0$
- (iii) $\sum_{k=0}^{p-1} X_k = 0$
- (iv) $\sum_{k=0}^{p-1} |X_k|^2 = \frac{1}{2}$

Show that the signal x can be expressed in the form $e^{i(An+B)} \sin(Cn)$, and determine what the parameters A , B , and C are. There may be multiple possible sets of parameters that satisfy the conditions.

Hint: Start by finding the signal $x[n]$ using the synthesis equation. Then you should use Reverse Euler's Formula to separate out a $\sin(Cn)$ term from the equation.

Solution:

From hint (i), we can see that $x[n]$ has a period of 3. Therefore, we can plug in $p = 3$ and $\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{3}$ for the rest of our hints.

Recall the analysis equation

$$X_k = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^{-ik\omega_0 n}$$

With our analysis equation, we can solve X_0 as

$$X_0 = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^{-i(0)\frac{2\pi}{3}n} = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^0 = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] = \frac{1}{p_x} (0) = 0$$

From here, we notice that

$$\sum_{k=0}^{p-1} X_k = \sum_{k=0}^{3-1} X_k = X_0 + X_1 + X_2 = 0$$

Since $X_0 = 0$, we have $X_0 + X_1 + X_2 = 0 + X_1 + X_2 = 0$, which means that $X_1 = -X_2$.

Finally, we can expand the last hint as follows:

$$\begin{aligned}
 \sum_{k=0}^{p-1} |X_k|^2 &= \frac{1}{2} \\
 |X_0|^2 + |X_1|^2 + |X_2|^2 &= \frac{1}{2} \\
 |0|^2 + |-X_2|^2 + |X_2|^2 &= \frac{1}{2} \\
 0 + |X_2|^2 + |X_2|^2 &= \frac{1}{2} \\
 2|X_2|^2 &= \frac{1}{2} \\
 |X_2|^2 &= \frac{1}{4}
 \end{aligned}$$

From here there are many possible values for X_2 we could choose (an infinite number, actually), but the simplest is $X_2 = \frac{1}{2}$.

At this point, we have that $X_0 = 0$, $X_1 = -X_2 = -\frac{1}{2}$, and $X_2 = \frac{1}{2}$. Plugging these into our synthesis equation, we get:

$$\begin{aligned}
 x[n] &= \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \\
 &= \sum_{k=0}^2 X_k e^{ik\frac{2\pi}{3}n} \\
 &= X_0 e^{i(0)\frac{2\pi}{3}n} + X_1 e^{i(1)\frac{2\pi}{3}n} + X_2 e^{i(2)\frac{2\pi}{3}n} \\
 &= (0) e^{i(0)\frac{2\pi}{3}n} + \left(-\frac{1}{2}\right) e^{i(1)\frac{2\pi}{3}n} + \left(\frac{1}{2}\right) e^{i(2)\frac{2\pi}{3}n} \\
 &= -\frac{1}{2} e^{i\frac{2\pi}{3}n} + \frac{1}{2} e^{i\frac{4\pi}{3}n} \\
 &= \frac{1}{2} e^{(i\pi n + i\frac{\pi}{3}n)} - \frac{1}{2} e^{(i\pi n - i\frac{\pi}{3}n)} \\
 &= \frac{1}{2} e^{i\pi n} (e^{i\frac{\pi}{3}n} - e^{-i\frac{\pi}{3}n}) \\
 &= e^{i\pi n} i \frac{(e^{i\frac{\pi}{3}n} - e^{-i\frac{\pi}{3}n})}{2i} \\
 &= e^{i\pi n} e^{i\frac{\pi}{2}} \sin\left(\frac{\pi}{3}n\right) \\
 &= e^{i(\pi n + \frac{\pi}{2})} \sin\left(\frac{\pi}{3}n\right)
 \end{aligned}$$

Pattern matching, we can see that $A = \pi$, $B = \frac{\pi}{2}$, and $C = \frac{\pi}{3}$ are parameters such that $x[n] = e^{i(An+B)} \sin(Cn)$ satisfies all of our described qualities.

4. Parseval's Theorem

Let x be a **real valued** periodic signal that repeats every p samples. Additionally, let X be the DTFS of x . Show that the following is true:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{k=0}^{p-1} x[k]^2$$

To do this, we will first prove something that is tangentially related to the above problem, and use it to prove the above.

- (a) Let $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthogonal basis that spans $\mathbb{C}^n$, where each vector $\mathbf{v}_i$ has norm m . Given that a vector $\mathbf{y} \in \mathbb{R}^n$ can be expressed as:

$$\mathbf{y} = b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n = y_1 \mathbf{e}_1 + \dots + y_n \mathbf{e}_n = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

What is $\langle \mathbf{y}, \mathbf{y} \rangle$ in terms of $y_1, \dots, y_n$? What about in terms of $b_1, \dots, b_n$?

Solution: We can trivially express $\langle \mathbf{y}, \mathbf{y} \rangle$ in terms of $y_1, \dots, y_n$ as $y_1^2 + y_2^2 + \dots + y_n^2 = \sum_{k=1}^n y_k^2$, since $\langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{y}\|^2$.

We can also express $\langle \mathbf{y}, \mathbf{y} \rangle$ in basis V as follows:

$$\langle \mathbf{y}, \mathbf{y} \rangle = \langle b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n, b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n \rangle$$

By properties of complex inner products and the fact that $\langle \mathbf{v}_j, \mathbf{v}_k \rangle = 0$ for $j \neq k$, we can equate this to the following:

$$\langle b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n, b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n \rangle = b_1 b_1^* \langle \mathbf{v}_1, \mathbf{v}_1 \rangle + \dots + b_n b_n^* \langle \mathbf{v}_n, \mathbf{v}_n \rangle$$

Since $\langle \mathbf{v}_j, \mathbf{v}_j \rangle = \|\mathbf{v}_j\|^2 = m^2$, and $b_j b_j^* = |b_j|^2$, we can simplify this further:

$$b_1 b_1^* \langle \mathbf{v}_1, \mathbf{v}_1 \rangle + \dots + b_n b_n^* \langle \mathbf{v}_n, \mathbf{v}_n \rangle = m^2 |b_1|^2 + \dots + m^2 |b_n|^2 = m^2 \cdot \sum_{k=1}^n |b_k|^2$$

Thus,

$$\langle \mathbf{y}, \mathbf{y} \rangle = \sum_{k=1}^n y_k^2 = m^2 \cdot \sum_{k=1}^n |b_k|^2$$

- (b) Using what we have learned above, show that:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{k=0}^{p-1} x[k]^2$$

Solution: Let the vector representation of our signal x be $\mathbf{x} = \begin{bmatrix} x[0] \\ \vdots \\ x[p-1] \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$, to mirror the

logic of part (a). We know that DTFS is an orthogonal basis that spans $\mathbb{C}^p$ where each basis vector has norm $\sqrt{p}$. Thus, we know that expressing x in the DTFS basis would give us coefficients $b_1, \dots, b_p$ that satisfy what we showed in part (a). By definition of DTFS, we know that these coefficients are $X_0, \dots, X_{p-1} = b_1, \dots, b_p$. Plugging these into what we showed from part (a):

$$\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{k=1}^p x_k^2 = (\sqrt{p})^2 \cdot \sum_{k=1}^p |b_k|^2 \rightarrow \sum_{k=1}^p x[k]^2 = p \cdot \sum_{k=1}^p |X_k|^2$$

As mentioned in the problem title, this is an instance of Parseval's theorem!

- (c) Let Y be the DFT representation of x . Recall that the DFT has the DTFS basis vectors scaled by $\frac{1}{p}$. What is $\sum_{k=0}^{p-1} |Y_k|^2$ in terms of $x[0], \dots, x[p-1]$?

Solution:

Following the logic of part (b), let the vector representation of our signal x be $\mathbf{x} = \begin{bmatrix} x[0] \\ \dots \\ x[p-1] \end{bmatrix} = \begin{bmatrix} x_1 \\ \dots \\ x_p \end{bmatrix}$.

We know that DFT is an orthogonal basis that spans $\mathbb{C}^p$. As such, DFT is an orthogonal basis where each basis vector has norm $\sqrt{p}/p = 1/\sqrt{p}$. Thus, we know that expressing x in the DFT basis would give us coefficients $b_1, \dots, b_p$ that satisfy what we showed in part (a). By definition of DFT, we know that these coefficients are $Y_0, \dots, Y_{p-1} = b_1, \dots, b_p$. Plugging these into what we showed from part (a):

$$\frac{1}{p} \sum_{k=0}^{p-1} |Y_k|^2 = \sum_{k=0}^{p-1} x[k]^2$$

Note the $\frac{1}{p}$ in front of the summation, which appears because the norm of each basis vector is $\frac{1}{\sqrt{p}}$ rather than $\sqrt{p}$ as it was in the DTFS basis.

5. Signal Properties

We are given that a p -periodic, real valued time signal $x[n]$ is odd; that is, $x[n] = -x[-n], \forall n \in \mathbb{Z}$.

Let $\{X_k | k \in \langle p \rangle\}$ be the set of DTFS coefficients of x . $\langle p \rangle$ denotes a suitable discrete interval of length p (i.e., an interval containing p contiguous integers). For example, $\sum_{k \in \langle p \rangle} X_k$ may denote $\sum_{k=0}^{p-1} X_k$ or $\sum_{k=1}^p X_k$.

- (a) Show that X_k is odd in k . More formally, show that $X_{-k} = -X_k, \forall k \in \langle p \rangle$

Hint: Think of a suitable change of variables for n in your analysis equation.

Solution: We begin by writing the analysis equation for X_{-k} :

$$X_{-k} = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{ik\omega_0 n}$$

Let's use a change of variables $m = -n$. Note that inverting each n still produces a suitable discrete interval of length p .

$$\implies X_{-k} = \frac{1}{p} \sum_{m \in \langle p \rangle} x[-m] e^{-ik\omega_0 m}$$

Using the fact that $x[-m] = -x[m]$:

$$\implies X_{-k} = -\frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} = -X_k$$

- (b) Show that X_k is purely imaginary $\forall k \in \langle p \rangle$, and has no real component.

Hint: To show that a complex number z is purely imaginary, show that $z^ = -z$.*

Solution:

$$X_k = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}$$

$$\implies X_k^* = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{ik\omega_0 n}$$

We can proceed exactly as we did in part (a), by using a change of variables $m = -n$.

$$\implies X_k^* = \frac{1}{p} \sum_{m \in \langle p \rangle} x[-m] e^{-ik\omega_0 m}$$

Using the fact that $x[-m] = -x[m]$:

$$\implies X_k^* = -\frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} = -X_k$$

Now, we are given that a p -periodic, real-valued time signal $y[n]$ is even; that is, $y[n] = y[-n], \forall n \in \mathbb{Z}$.

Let $\{Y_k | k \in \langle p \rangle\}$ be the set of DTFS coefficients of y .

- (c) Show that Y_k is purely real $\forall k \in \langle p \rangle$, and has no imaginary component.

Solution: To show a complex number z is purely real, show that $z^* = z$.

$$Y_k = \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{-ik\omega_0 n}$$

$$\implies Y_k^* = \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{ik\omega_0 n}$$

Again, let $m = -n$:

$$\implies Y_k^* = \frac{1}{p} \sum_{m \in \langle p \rangle} y[-m] e^{-ik\omega_0 m}$$

Since $y[m] = y[-m]$:

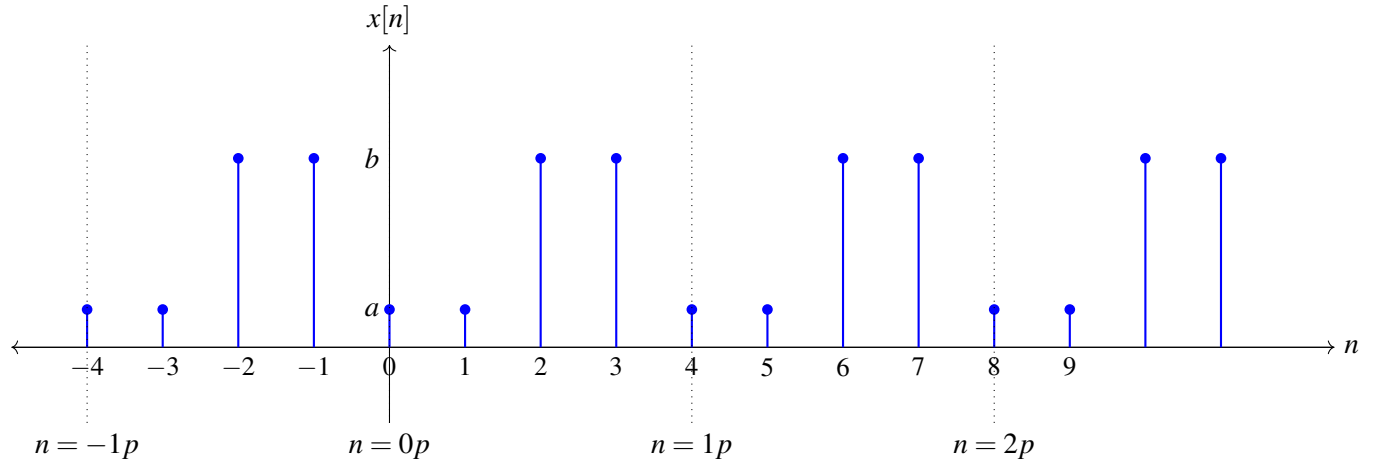
$$= Y_k^* = \frac{1}{p} \sum_{m \in \langle p \rangle} y[m] e^{-ik\omega_0 m} = Y_k$$

6. "Square" Waves

You are given a p -periodic "square" wave signal x defined for its first period as follows, where p is even:

$$x[n] = \begin{cases} a & \text{if } 0 \leq n < p/2 \\ b & \text{if } p/2 \leq n \leq p-1 \end{cases}$$

To help you understand the notation above, below is a graph of the given signal, for $p = 4$.

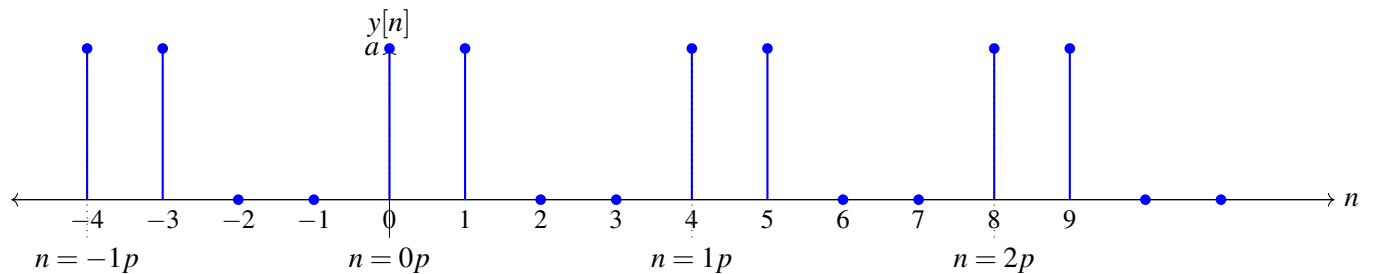


- (a) Let X_k represent the DTFS coefficients of x . Show that $X_0 = \frac{a+b}{2}$.

Solution:

$$\begin{aligned} X_0 &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-i0\omega_0 n} = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] = \frac{1}{p} \sum_{n=0}^{p-1} x[n] \\ &= \frac{1}{p} \left(\sum_{n=0}^{p/2-1} x[n] + \sum_{n=p/2}^{p-1} x[n] \right) = \frac{1}{p} \left(a \cdot \frac{p}{2} + b \cdot \frac{p}{2} \right) = \frac{a}{2} + \frac{b}{2} = \frac{a+b}{2} \end{aligned}$$

- (b) Now assume that y is a signal defined the same as x , except the value of b is 0. For clarity, the case for which $p = 4$ is shown below:



What are the DTFS coefficients Y_k of this signal y with fundamental period equal to p ? Your answer should be in terms of k , a , and p and should include a summation.

Solution: We can apply our analysis equation in order to solve for the Fourier coefficients:

$$Y_k = \frac{1}{p} \sum_{n=0}^{p-1} y[n] e^{-ik\omega_0 n}$$

Notice that for $\frac{p}{2} \leq n \leq p-1$, $y[n]$ is equal to 0, so those terms in the summation disappear.

$$Y_k = \frac{1}{p} \sum_{n=0}^{p/2-1} y[n] e^{-ik\omega_0 n}$$

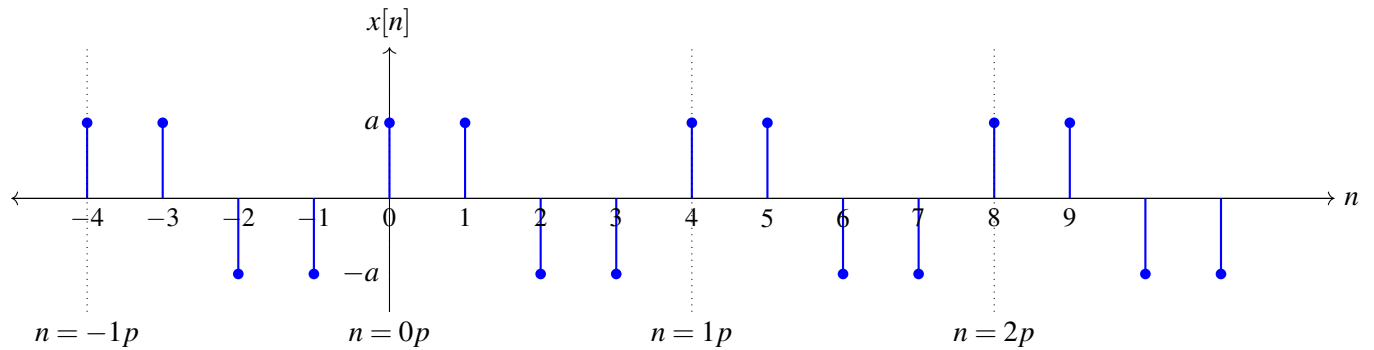
For $0 \leq n \leq \frac{p}{2} - 1$, $y[n]$ is equal to a .

$$Y_k = \frac{1}{p} \sum_{n=0}^{p/2-1} a e^{-ik\omega_0 n} = \frac{a}{p} \sum_{n=0}^{p/2-1} e^{-ik\omega_0 n}$$

For the final step, we can express ω_0 in terms of p to express Y_k in the desired terms:

$$Y_k = \frac{a}{p} \sum_{n=0}^{p/2-1} e^{-ik \frac{2\pi}{p} n}$$

- (c) Now assume that z is a signal defined the same as x , except the value of b is now equal to $-a$. This signal is now "half-wave symmetric." Again, the case in which $p = 4$ is shown below:



Express the DTFS coefficients Z_k in terms of the Fourier coefficients Y_k . Your answer should be in terms of k and Y_k .

Hint: Recognize $z[n]$ as the sum of two signals and use the time shift property.

Solution: Following the hint, we notice that $z[n]$ is $y[n]$ minus another $y[n]$ shifted by $\frac{p}{2}$. Mathematically:

$$z[n] = y[n] - y\left[n + \frac{p}{2}\right]$$

Let $\hat{Y}_k$ represent the DTFS coefficients of $y\left[n + \frac{p}{2}\right]$. Due to the linearity of DTFS, we know that

$$Z_k = Y_k - \hat{Y}_k$$

Applying the time shift property,

$$\hat{Y}_k = e^{-ik\omega_0 \frac{p}{2}} Y_k = e^{ik \frac{2\pi}{p} \frac{p}{2}} Y_k = e^{ik\pi} Y_k$$

$$Z_k = Y_k - e^{ik\pi} Y_k$$

$$Z_k = Y_k \left(1 - e^{ik\pi}\right) = Y_k \left(1 - (-1)^k\right)$$

(d) Using your result to part (c), show that given any even $k \in \langle p \rangle$, $Z_k = 0$.

Solution: If k is even, it is a multiple of 2, so let $k = 2c$, where c is an integer. The result from part (c) becomes

$$Z_k = Y_k \left(1 - e^{i(2c)\pi} \right) = Y_k \left(1 - (e^{i2\pi})^c \right) = Y_k (1 - 1^c) = Y_k (1 - 1) = 0$$

In fact, for any half-wave symmetric waveform, the even Fourier coefficients are 0; the steps we did in parts (c) and (d) didn't rely on the fact that our signal was a square wave!

7. 2-D DTFS

In this problem, you will implement and use the 2-D Discrete-Time Fourier Series (DTFS) on small images and then reconstruct the images using some of their DTFS coefficients. Use Python for all coding components.

Images can be thought of as signals too! For an image $x \in \mathbb{R}^{N \times M}$, we can think of it as a signal $x \in \mathbb{R}^{N \times M}$. For a signal $x \in \mathbb{R}^{N \times M}$, the complex-exponential basis is indexed by (k, ℓ) and given by

$$\psi_{k,\ell}[n, m] = e^{i2\pi\left(\frac{kn}{N} + \frac{\ell m}{M}\right)}, \quad n = 0, \dots, N-1, m = 0, \dots, M-1.$$

These basis vectors are orthogonal under the inner product on $\mathbb{C}^{N \times M}$ and yield the 2-D DTFS analysis and synthesis equations:

$$X[k, \ell] = \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} x[n, m] e^{-i2\pi\left(\frac{kn}{N} + \frac{\ell m}{M}\right)}, \quad x[n, m] = \sum_{k=0}^{N-1} \sum_{\ell=0}^{M-1} X[k, \ell] e^{i2\pi\left(\frac{kn}{N} + \frac{\ell m}{M}\right)}.$$

This problem has an IPython component. Please attach a pdf printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

- (a) Let the image be $x \in \mathbb{R}^{N \times M}$. Implement the 2-D DTFS and its inverse from the definition above. To test your implementation, use the provided 32×32 images (horizontal stripes, vertical stripes, diagonal stripes, checkerboard, and dots). Read each image into an array x (treat the pixels as real-valued), compute X using your implementation, and reconstruct $\hat{x}$ from X . Report the maximum absolute error $\max_{n,m} |\hat{x}[n, m] - x[n, m]|$ for each image, between the original and reconstructed images.

Solution: Implement the definitions exactly as written. Test correctness by confirming the reconstruction error is close to zero. See the solution IPython notebook for an example of a correct implementation.

- (b) Visualize several 2-D DTFS basis components to understand how (k, ℓ) jointly determine the wave pattern. For a fixed $N = M = 32$, generate and display $\Re\{\psi_{k,\ell}\}$ for $k, \ell \in \{0, 3, 5\}$. Briefly describe how changing k versus ℓ affects the horizontal and vertical oscillations.

Solution: Increasing k increases vertical oscillations while increasing ℓ increases horizontal oscillations. For nonzero k and ℓ , the pattern mixes both directions. Wave patterns are along the 45 degree diagonal when $k = \ell$. See the solution IPython notebook for an example of a correct implementation of the basis functions.

- (c) Verify orthogonality of the complex exponential basis functions in 2-D. Specifically, numerically evaluate

$$\left\langle e^{i2\pi\left(\frac{kn}{N} + \frac{\ell m}{M}\right)}, e^{i2\pi\left(\frac{k'n}{N} + \frac{\ell'm}{M}\right)} \right\rangle = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} e^{i2\pi\left(\frac{kn}{N} + \frac{\ell m}{M}\right)} e^{-i2\pi\left(\frac{k'n}{N} + \frac{\ell'm}{M}\right)},$$

for several distinct index pairs $(k, \ell) \neq (k', \ell')$, and also the self-inner product when $(k, \ell) = (k', \ell')$.

Solution: You should observe 0 (up to floating-point error) when $(k, \ell) \neq (k', \ell')$ and NM when $(k, \ell) = (k', \ell')$. See the solution IPython notebook for an example of a correct implementation of the inner product.

- (d) Compute both expressions below and report the difference between them.

$$\sum_{n=0}^{N-1} \sum_{m=0}^{M-1} |x[n, m]|^2 \quad \text{vs.} \quad NM \sum_{k=0}^{N-1} \sum_{\ell=0}^{M-1} |X[k, \ell]|^2.$$

You should find that both expressions are equal (within machine precision). This result is known as Parseval's theorem.

Solution: Compute both quantities from your implementations; the relative error should be near machine precision for the test patterns. See the solution IPython notebook for an example of a correct implementation of both sides of the equation.

- (e) For each provided image, plot the magnitude heatmap $|X[k, \ell]|$ (with labeled axes). Provide a brief description of what you see in each heatmap. The code for this part is provided.

Solution: Horizontal stripes produce a sequence of spikes with $\ell = 0$, vertical stripes produce a sequence of spikes with $k = 0$, diagonal stripes produce spikes along $k = \ell$, checkerboard produces a grid of varying-intensity spikes, and dots in time domain yield a grid of spikes in the frequency domain. See the solution IPython notebook for example heatmaps. Note that every signal has a DC offset ($k=0, \ell=0$) because the pixel values are always between 0 and 1.

- (f) Consider the checkerboard image x with $N = M = 32$ and its exact set of DTFS coefficients X . In this part, we will be reconstructing the image from a subset of its DTFS coefficients. Use each of the following three index-selection rules:

- (i) Square set: keep indices with $k \leq \sqrt{B}$ and $\ell \leq \sqrt{B}$ for a coefficient budget of B .
- (ii) Quarter circular set: keep indices with $k^2 + \ell^2 \leq R^2$.
- (iii) Max set: keep the M largest magnitude coefficients.

For three budgets: 64, 256, and 576 **total** kept coefficients, reconstruct $\hat{x}$ for each rule and compute the mean-squared error (MSE) in the image reconstruction. Provide the reconstructions for each rule and budget. Briefly discuss any visual reconstruction errors and compare the three rules. **Note:** For the quarter circular set, it's fine if the number of kept coefficients is slightly different from the budget.

Solution: The reconstruction quality improves as more coefficients are included. The max set has perfect approximation behavior at just 64 coefficients. This is because the checkerboard DTFS only has nonzero coefficients for 17 values of (k, ℓ) . The square set and quarter circular set seem to perform similarly, but both display substantial edge effects at the boundaries of the image, due to lack of inclusion of the high-frequency coefficients (which allow for sharp edges).

8. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.